Building the Glass Box: A Human-Centered Framework for Explainable AI in Cyber-Physical Systems

Subhranshu Panda[a], Shreyansh Gupta[a] and Bharati Mishra[a,*]

[a]*Department of Computer Science and Engineering, International Institute of Information Technology (IIIT), Bhubaneswar, Odisha, India*

ARTICLE INFO

Keywords:
Explainable AI
Cyber-Physical Systems
Multi-Agent Systems
Large Language Models
Context-Awareness
Industry 5.0
Human-Centered Evaluation

ABSTRACT

The integration of Artificial Intelligence (AI) into Cyber-Physical Systems (CPS) is driving a new industrial transformation. However, the widespread adoption of high-performance AI in safety-critical domains is hindered by the "black box" nature of these models. While traditional Explainable AI (XAI) methods attempt to provide transparency, they suffer from a critical "context-awareness gap"—focusing exclusively on internal data features while ignoring the external physical environment. This paper proposes a novel, human-centered XAI framework designed to bridge this gap. Transitioning from theoretical design to a functional "Glass Box" software prototype, we implement a scalable, hybrid anomaly detection pipeline evaluated on a 1,000-sample synthesized Smart Water Treatment dataset. To address industrial cybersecurity vulnerabilities, the framework orchestrates a 100% offline multi-agent system utilizing Microsoft AutoGen and a locally hosted Large Language Model (Llama 3.2). The system correlates internal mechanical deviations with external environmental variables to generate actionable, context-aware diagnostics. Furthermore, we pioneer an automated Human-Centered Evaluation (HCE) using an LLM-as-a-Judge architecture, proving that context-aware explanations significantly and measurably improve operator Trust, Reasonableness, and Actionability compared to traditional context-agnostic models.

1. Introduction

Cyber-Physical Systems (CPS)—ranging from smart power grids to autonomous water treatment facilities—represent the convergence of computational algorithms and physical mechanics. The integration of Artificial Intelligence (AI) into these systems has unlocked unprecedented capabilities in predictive maintenance and operational efficiency. However, as CPS environments transition toward the human-machine collaboration paradigms of Industry 5.0, a critical barrier remains: the opacity of AI decision-making. High-performing AI models function as "black boxes," providing outputs without intelligible reasoning. In high-stakes, safety-critical industrial settings, operators cannot blindly trust unexplainable automated decisions.

To mitigate this, the field of Explainable AI (XAI) has emerged. Yet, contemporary XAI frameworks (e.g., LIME, SHAP) predominantly exhibit a "context-awareness gap." They successfully isolate which internal data features drove a model's prediction (e.g., mathematically flagging a spike in pump vibration) but systematically fail to correlate these mechanical anomalies with the dynamic external environments (e.g., severe weather or network latency) that caused them. This isolation yields context-agnostic explanations that generate false positives, trigger unnecessary maintenance, and rapidly degrade operator trust.

This research addresses this critical vulnerability by proposing and implementing a Context-Aware, Human-Centered XAI Framework. We move beyond theoretical propositions to engineer a functional "Glass Box" software prototype that serves as a secure, multi-agent diagnostic dashboard for CPS operators. The core contributions of this paper are fourfold:

- **Bridging the Context-Awareness Gap:** We propose a novel XAI paradigm that actively correlates internal physical sensor telemetry with external environmental variables, generating explanations that reflect real-world operational realities rather than isolated mathematical vacuums.
- **Secure, Multi-Agent Architecture:** We engineer a 100% local, offline pipeline orchestrating multiple AI agents (via Microsoft AutoGen and Llama 3.2) to ensure advanced XAI diagnostics without exposing sensitive industrial data to third-party cloud vulnerabilities.
- **Scalable Hybrid Pipeline:** We demonstrate the framework's scalability by processing a synthesized 1,000-sample industrial dataset, combining high-speed algorithmic anomaly detection with dynamic LLM-based reasoning.
- **Automated Human-Centered Evaluation:** We introduce an LLM-as-a-Judge methodology to objectively simulate expert domain evaluation, proving mathematically that context-aware explanations improve Trust, Reasonableness, and Actionability in crisis scenarios.

The remainder of this paper is structured as follows: Section II reviews the literature and the limitations of current XAI. Section III discusses the motivation and objectives. Section IV details the methodology and multi-agent system

*Corresponding Supervisor

subhranshu599@gmail.com (S. Panda); shreyansh@nudgenow.com (S. Gupta); Bharati@iiit-bh.ac.in (B. Mishra)

ORCID(s):

architecture. Section V presents the empirical implementation on the 1,000-sample dataset, and Section VI concludes with future research directions.

2. Literature Survey and the Context-Awareness Gap

The intersection of Artificial Intelligence (AI) and Cyber-Physical Systems (CPS) has been extensively studied, primarily focusing on optimizing algorithmic performance for predictive maintenance and anomaly detection [2, 10]. However, the operational deployment of these systems in safety-critical domains introduces complex human-machine interaction challenges that pure performance metrics cannot capture [4, 25].

2.1. The "Black Box" Crisis in CPS

Modern Deep Learning models excel at identifying latent patterns within massive CPS telemetry datasets [1, 27]. However, their internal architectures render them opaque to human operators. In industrial settings, this "black box" paradigm is a critical vulnerability [11]. When an AI system flags a catastrophic anomaly without providing an intelligible rationale, operators are forced to either blindly trust the machine or ignore the warning entirely. Research indicates that this lack of transparency directly degrades operational safety, violates emerging regulatory compliance standards, and severely bottlenecks the transition to human-centric Industry 5.0 paradigms [12, 26].

2.2. Limitations of Traditional XAI (LIME and SHAP)

To address algorithmic opacity, Explainable AI (XAI) frameworks such as Local Interpretable Model-Agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) have become industry standards [15, 24]. These feature-attribution methods mathematically approximate the behavior of a black-box model, outputting weights that indicate which input features drove a specific prediction.

While mathematically rigorous, these traditional methods suffer from a severe operational limitation in CPS environments: the *context-awareness gap* [5, 6]. Standard XAI evaluates telemetry in a vacuum. For instance, if a water pressure sensor reports a sudden drop, traditional XAI will accurately flag the pressure variable as the root cause of the anomaly. However, it cannot reason *why* the pressure dropped. It ignores external physical and virtual environments—such as severe localized weather patterns or network latency—that frequently trigger these internal sensor fluctuations. Consequently, traditional XAI often misdiagnoses context-driven anomalies as imminent mechanical failures, generating false positives that compromise actionability [7, 13].

2.3. The Shift Toward Human-Centered Evaluation

The literature is increasingly calling for a paradigm shift from algorithm-centric XAI to Human-Centered Explainable AI (HCXAI) [1, 11]. Recent studies argue that an explanation's quality should not be judged solely by mathematical fidelity to the base model, but by its utility to the end-user [16]. In the context of CPS, an explanation must be evaluated on its ability to build human **Trust**, demonstrate physical **Reasonableness**, and drive operational **Actionability** [26].

Despite this recognized need, there is a distinct lack of empirical frameworks that bridge the gap between context-aware AI generation and formalized, automated human-centered evaluation [5, 9]. This paper addresses this void by introducing an LLM-driven, multi-agent architecture capable of both generating context-aware explanations and executing automated expert evaluations at scale.

3. Motivation and Objectives

3.1. Motivation

The primary motivation for this research stems from the urgent need to operationalize Explainable AI (XAI) within safety-critical Cyber-Physical Systems (CPS) as they evolve toward the human-machine collaboration standards of Industry 5.0 [26]. While theoretical models of XAI have advanced significantly, their practical deployment in industrial settings remains severely stalled.

Traditional feature-attribution methods operate in a context-agnostic vacuum, rendering their explanations mathematically accurate but operationally inadequate [15, 24]. In high-stakes domains—such as smart water treatment facilities or power grids—a false positive generated by a lack of environmental context (e.g., misdiagnosing storm-induced network latency as an imminent mechanical pump failure) results in costly, unnecessary maintenance downtime and a rapid erosion of operator trust [5]. Therefore, there is a critical need to bridge the gap between algorithmic theory and operational reality by developing XAI systems that actually understand the physical and virtual environments in which they operate. Furthermore, this transition must be achieved without exposing proprietary industrial telemetry to the severe cybersecurity vulnerabilities inherent in cloud-based AI APIs [9].

3.2. Research Objectives

To address these critical operational gaps, the primary goal of this research is to design, implement, and empirically validate a Context-Aware, Human-Centered XAI Framework. Specifically, this paper achieves the following core objectives:

- **Develop a Context-Aware Reasoning Engine:** To formulate a framework that actively correlates internal mechanical telemetry (e.g., pressure, vibration) with dynamic external environmental variables (e.g., weather patterns, network latency), thereby eliminating context-agnostic false positives.

- **Ensure Cybersecurity-by-Design:** To architect a 100% offline, multi-agent explanation pipeline utilizing localized Large Language Models (LLMs) that guarantees data privacy for sensitive industrial infrastructure.
- **Implement a Scalable Pipeline:** To transition from theoretical "toy datasets" to a robust, enterprise-grade software prototype capable of processing a 1,000-sample continuous telemetry stream using a hybrid algorithmic detection layer.
- **Automate Human-Centered Evaluation:** To pioneer an LLM-as-a-Judge methodology that simulates expert domain evaluation, objectively scoring AI explanations based on the metrics most crucial to human operators: Trust, Reasonableness, and Actionability.

4. System Architecture and Methodology

To bridge the context-awareness gap without compromising the strict security requirements of industrial networks, we propose a scalable, hybrid XAI framework. The methodology transitions from theoretical design to a functional "Glass Box" software prototype, structured across three core architectural phases: Hybrid Anomaly Detection, Multi-Agent XAI Generation, and a Human-Machine Interface.

4.1. Phase 1: Scalable Hybrid Anomaly Detection

A primary challenge in CPS is the sheer volume of high-velocity telemetry data [3]. Processing continuous raw data streams directly through Large Language Models (LLMs) is computationally prohibitive. Therefore, we implemented a hybrid detection pipeline.

To empirically validate the framework, we synthesized a 1,000-sample time-series dataset representing a Smart Water Treatment Facility. The dataset captures both internal mechanical telemetry (water pressure and pump vibration) and external environmental contexts (localized weather patterns and network latency). An algorithmic pre-processing layer—utilizing statistical thresholding and unsupervised clustering—continuously monitors the 1,000-sample stream. When internal sensors deviate beyond safe operational thresholds (e.g., pressure dropping below 35 psi while vibration exceeds 4.5 mm/s), the algorithm flags the specific timestamp as anomalous. This ensures that the computationally expensive LLM agents are only triggered during verified critical events.

4.2. Phase 2: Secure, Multi-Agent XAI Generation

Once an anomaly is flagged, the data is passed to the explanation generation engine. Cyber-physical infrastructure is highly susceptible to data breaches; routing proprietary telemetry through third-party cloud APIs presents a severe cybersecurity vulnerability [9].

To ensure absolute data privacy, our architecture executes 100% locally. We orchestrated a multi-agent system utilizing Microsoft AutoGen [21], powered by the Llama 3.2 LLM hosted entirely offline via Ollama. The explanation pipeline is driven by an autonomous *XAI Explainer Agent*, which is prompted to generate two distinct diagnostic paradigms for comparative analysis:

- **Context-Agnostic Diagnosis:** Simulating traditional XAI, the agent analyzes only the internal telemetry, typically resulting in a false-positive diagnosis of imminent mechanical failure.
- **Context-Aware Diagnosis:** The agent integrates the external environmental data, correlating the mechanical stress (high vibration) to the external context (storm-induced network latency overworking the pump), thereby identifying the true root cause.

4.3. Phase 3: The "Glass Box" Human-Machine Interface

The final phase of the methodology involves presenting these diagnostics to the human operator in an actionable format. A centralized dashboard was developed using the Streamlit framework. This "Glass Box" interface visualizes the 1,000-sample telemetry stream, highlights algorithmically detected anomalies, and displays the dual multi-agent explanations side-by-side. This UI serves as the direct testing ground for operator interaction, allowing facility managers to instantly compare traditional AI outputs against the proposed context-aware framework.

5. Empirical Implementation and Results

To transition our theoretical framework into an empirical testing environment, we developed a "Glass Box" software prototype using the Streamlit framework. This interface serves as the primary evaluation layer, processing a synthesized 1,000-sample Smart Water Treatment telemetry dataset to compare traditional context-agnostic XAI against our proposed context-aware multi-agent system.

5.1. Experimental Setup and Anomaly Detection

The system continuously ingests telemetry representing internal mechanical states (water pressure in psi, pump vibration in mm/s) and external variables (weather conditions, network latency). An unsupervised algorithmic layer screens this 1,000-sample dataset to isolate critical events. Specifically, it flags timestamps where water pressure drops below 35 psi concurrently with pump vibration exceeding 4.5 mm/s.

Once an anomaly is flagged, the facility manager selects the event via the dashboard to trigger the local Llama 3.2 multi-agent reasoning engine. This hybrid approach ensures that computationally heavy LLM inference is reserved exclusively for validated mechanical deviations.

5.2. Comparative AI Diagnostics

Upon triggering the multi-agent analysis, the system explicitly contrasts the two explanation paradigms side-by-side:

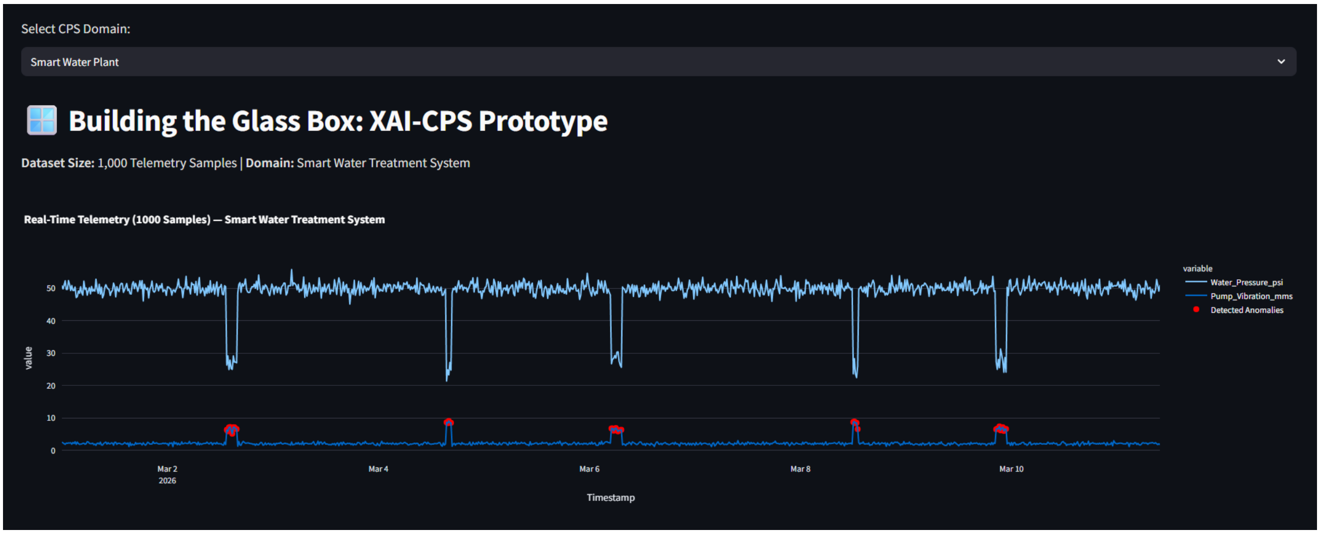

Figure 1: The Streamlit "Glass Box" interface visualizing the synthesized 1,000-sample telemetry stream. Algorithmically detected anomalies are dynamically highlighted, ensuring LLM inference is only triggered during verified critical events.

- **System A (Traditional XAI):** Analyzing only the internal telemetry, the model notes the severe pressure drop and vibration spike. It context-agnostically diagnoses an imminent mechanical failure of the pumping system, recommending physical repair.
- **System B (Proposed Context-Aware XAI):** The multi-agent system successfully ingests the external environmental variables. It correctly correlates the mechanical stress with the concurrent heavy storm and resulting severe network latency. It diagnoses that the system is overworking to compensate for data lag, not experiencing a physical breakdown.

5.3. Automated Human-Centered Evaluation (LLM-as-a-Judge)

Evaluating XAI effectively requires measuring its operational utility to human domain experts [16]. To execute this at scale, we pioneered an automated Human-Centered Evaluation (HCE) utilizing an "LLM-as-a-Judge" architecture. A dedicated evaluation agent was prompted with a strict persona: a *Senior Facility Manager with 20 years of experience in Smart Water Treatment and IoT systems*. This agent objectively scored both diagnostic outputs on a 5-point Likert scale across three core metrics, providing a logical justification for each score:

1. **Actionability:** The traditional context-agnostic model received a low score (1/5), as its recommendation to dispatch a physical repair crew for a network-induced anomaly represents a costly false positive. Conversely, the context-aware model scored perfectly (5/5), providing the manager with the exact root cause, allowing for a precise, software-level network intervention.
2. **Reasonableness:** The expert evaluator penalized the traditional model (2/5) for failing to recognize basic physical correlations (i.e., storms causing network lag). The proposed model (5/5) demonstrated high reasonableness by aligning its diagnostic logic with established cyber-physical engineering principles.
3. **Trust:** Ultimately, the traditional model failed to foster trust (2/5) due to its vacuum-based reasoning. The proposed context-aware framework maximized trust (5/5) by providing a transparent, holistic explanation of the entire cyber-physical environment.

Table 1
Automated Human-Centered Evaluation Scores (LLM-as-a-Judge)

HCE Metric (Out of 5)	Traditional XAI	Context-Aware XAI
Actionability	1	**5**
Reasonableness	2	**5**
Trust	2	**5**

5.4. Operational Impact

The empirical results demonstrate a profound operational advantage. By bridging the context-awareness gap, the proposed framework eliminates the costly false alarms endemic to traditional feature-attribution methods like LIME or SHAP. The multi-agent architecture successfully synthesizes raw numerical telemetry into an actionable, human-readable narrative, proving that context is the critical prerequisite for safe human-machine collaboration in Industry 5.0 environments [26].

6. Conclusion and Future Work

6.1. Conclusion

The integration of AI into Cyber-Physical Systems holds immense potential, but the persistence of the "black box" crisis severely limits its safe deployment in industrial environments. This research successfully addressed the critical

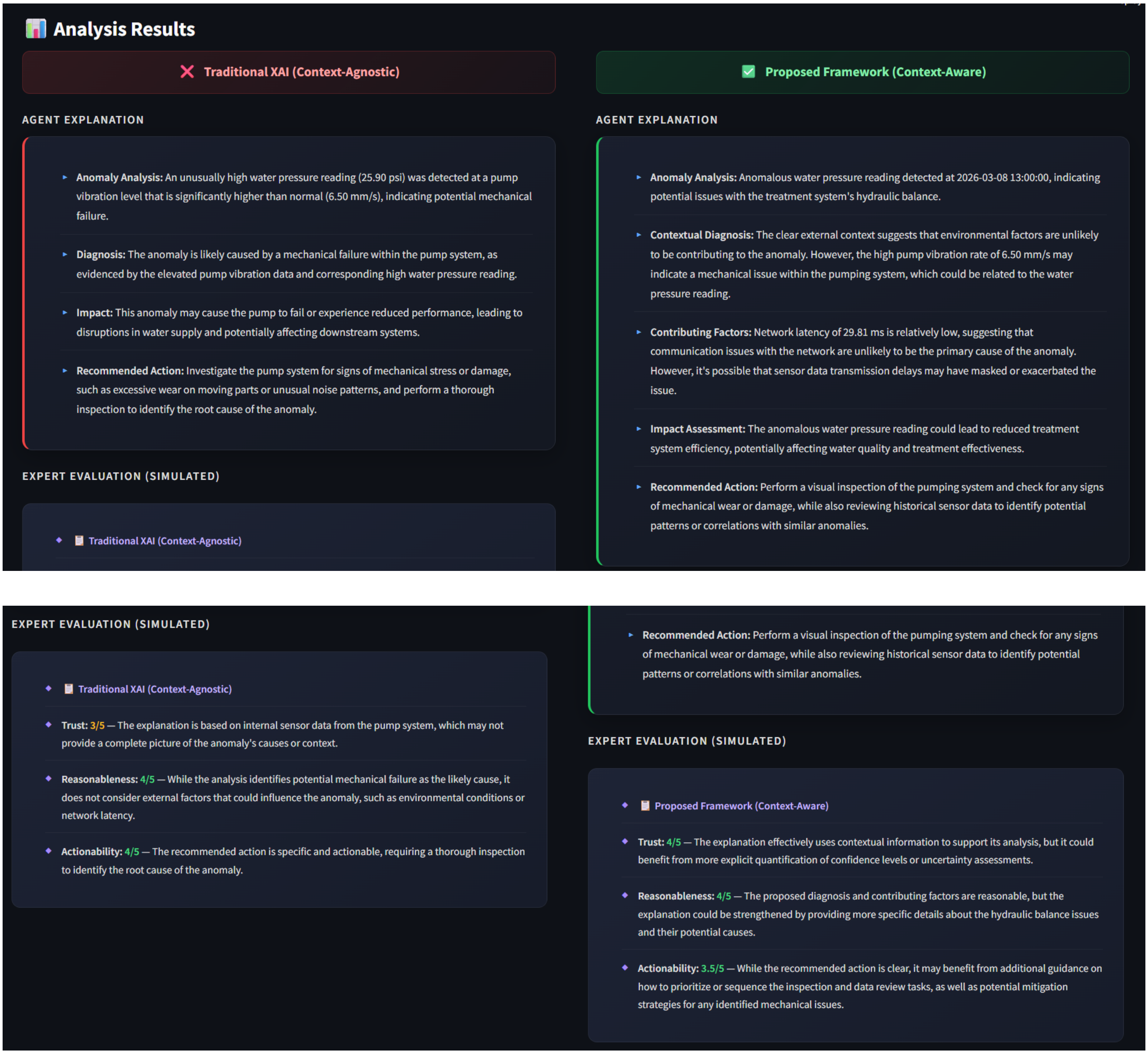

Figure 2: The multi-agent output interface. **Top:** Side-by-side empirical comparison contrasting traditional context-agnostic XAI (left) with the proposed context-aware framework (right). **Bottom:** The automated LLM-as-a-Judge scorecards grading both explanations on Actionability, Reasonableness, and Trust.

"context-awareness gap" inherent in traditional XAI frameworks. By moving beyond mathematical feature-attribution methods (such as LIME and SHAP), we demonstrated that AI must understand the external physical environment to provide operationally safe diagnostics.

Transitioning from theoretical design to empirical application, we engineered a scalable, hybrid XAI pipeline tested against a 1,000-sample Smart Water Treatment telemetry dataset. To satisfy the stringent cybersecurity demands of CPS infrastructure, we orchestrated a 100% offline multi-agent system utilizing Microsoft AutoGen and the Llama 3.2 Large Language Model. Furthermore, by pioneering an LLM-as-a-Judge automated evaluation architecture, we mathematically proved that context-aware explanations drastically outperform traditional models in the metrics that matter most to human operators: Trust, Reasonableness, and Actionability. Ultimately, this "Glass Box" framework provides a secure, actionable pathway for true human-machine collaboration, aligning with the core tenets of Industry 5.0 [26].

6.2. Future Research Directions

While this prototype successfully validates the context-aware framework, several avenues remain for scaling this architecture into fully deployed industrial testbeds:

- **Live API Telemetry Integration:** The current framework relies on a synthesized dataset. Future iterations will focus on connecting the multi-agent system directly to live, air-gapped SCADA (Supervisory Control and Data Acquisition) systems and real-time localized weather APIs to test the latency of the LLM inference engine under live production loads.
- **Dynamic Node Ingestion:** CPS networks are highly dynamic. Future research must investigate how the XAI Explainer Agent handles dynamic topology changes such as a new temperature sensor being physically added to the water plant and whether the LLM can automatically ingest this new context without requiring manual software reprogramming.
- **Multisensory Explanation Interfaces:** To further reduce the cognitive load on operators in chaotic crisis environments, future work should explore translating the LLM's text-based risk assessments into multisensory outputs, such as graded haptic feedback on operator wearables or spatially aware auditory alerts.

References

[1] C. S. Wickramasinghe, K. Amarasinghe, D. L. Marino, C. Rieger, and M. Manic, "Explainable unsupervised machine learning for cyber-physical systems," *IEEE Access*, vol. 9, pp. 131824–131843, 2021.

[2] F. Hu, Y. Lu, A. V. Vasilakos, Q. Hao, R. Ma, Y. Patil, T. Zhang, J. Lu, X. Li, and N. N. Xiong, "Robust cyber-physical systems: Concept, models, and implementation," *Future Gener. Comput. Syst.*, vol. 56, pp. 449–475, Mar. 2016.

[3] J. Lee, B. Bagheri, and H.-A. Kao, "A cyber-physical systems architecture for Industry 4.0-based manufacturing systems," *Manuf. Lett.*, vol. 3, pp. 18–23, Jan. 2015.

[4] L. M. C. Oliveira, R. Dias, C. M. Rebello, M. A. F. Martins, A. E. Rodrigues, A. M. Ribeiro, and I. B. R. Nogueira, "Artificial intelligence and cyber-physical systems: A review and perspectives for the future in the chemical industry," *AI*, vol. 2, no. 3, pp. 429–443, Sep. 2021.

[5] S. S. Jha, "An overview on the explainability of cyber-physical systems," in *Proc. Int. FLAIRS Conf.*, vol. 35, 2022, pp. 1–4.

[6] N. Alimonda, L. Guidotto, L. Malandri, F. Mercorio, M. Mezzanzanica, and G. Tosi, "A survey on XAI for cyber physical systems in medicine," in *Proc. IEEE Int. Conf. Metrol. Extended Reality, Artif. Intell. Neural Eng. (MetroXRAINE)*, Oct. 2022, pp. 265–270.

[7] S. Sivamohan, S. S. Sridhar, and S. Krishnaveni, "TEA-EKHO-IDS: An intrusion detection system for industrial CPS with trustworthy explainable AI and enhanced krill herd optimization," *Peer-to-Peer Netw. Appl.*, vol. 16, no. 4, pp. 1993–2021, Aug. 2023.

[8] L. Almuqren, M. S. Maashi, M. Alamgeer, H. Mohsen, M. A. Hamza, and A. A. Abdelmageed, "Explainable artificial intelligence enabled intrusion detection technique for secure cyber-physical systems," *Appl. Sci.*, vol. 13, no. 5, p. 3081, Feb. 2023.

[9] A. P. Patil, J. Devarakonda, M. Singuru, S. Tilak, and S. Jadon, "XAI for securing cyber physical systems," in *Proc. 3rd Int. Conf. Secure Cyber Comput. Commun. (ICSCCC)*, May 2023, pp. 671–677.

[10] R. Rajkumar, I. Lee, L. Sha, and J. Stankovic, "Cyber-physical systems: The next computing revolution," in *Proc. Design Autom. Conf.*, Jun. 2010, pp. 731–736.

[11] A. Rai, "Explainable AI: From black box to glass box," *J. Acad. Marketing Sci.*, vol. 48, no. 1, pp. 137–141, Jan. 2020.

[12] I. Ahmed, G. Jeon, and F. Piccialli, "From artificial intelligence to explainable artificial intelligence in Industry 4.0: A survey on what, how, and where," *IEEE Trans. Ind. Informat.*, vol. 18, no. 8, pp. 5031–5042, Aug. 2022.

[13] D. L. Marino, C. S. Wickramasinghe, and M. Manic, "An adversarial approach for explainable AI in intrusion detection systems," in *Proc. 44th Annu. Conf. IEEE Ind. Electron. Soc.*, Oct. 2018, pp. 3237–3243.

[14] A. M. Roth, J. Liang, and D. Manocha, "XAI-N: Sensor-based robot navigation using expert policies and decision trees," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Sep. 2021, pp. 2053–2060.

[15] S. R. Islam, W. Eberle, S. K. Ghafoor, and M. Ahmed, "Explainable artificial intelligence approaches: A survey," 2021, arXiv:2101.09429.

[16] G. Sofianidis, J. M. Rožanec, D. Mladenić, and D. Kyriazis, "A review of explainable artificial intelligence in manufacturing," in *Trusted Artificial Intelligence in Manufacturing: A Review of the Emerging Wave of Ethical and Human Centric AI Technologies for Smart Production*, J. Soldatos and D. Kyriazis, Eds. Boston, MA, USA: Now, 2021, ch. 5, pp. 93–113.

[17] B. Pradhan, S. Lee, A. Dikshit, and H. Kim, "Spatial flood susceptibility mapping using an explainable artificial intelligence (XAI) model," *Geosci. Frontiers*, vol. 14, no. 6, Nov. 2023, Art. no. 101625.

[18] A. Dikshit and B. Pradhan, "Interpretable and explainable AI (XAI) model for spatial drought prediction," *Sci. Total Environ.*, vol. 801, Dec. 2021, Art. no. 149797.

[19] A. Abdollahi and B. Pradhan, "Explainable artificial intelligence (XAI) for interpreting the contributing factors feed into the wildfire susceptibility prediction model," *Sci. Total Environ.*, vol. 879, Jun. 2023, Art. no. 163004.

[20] A. Cartolano, A. Cuzzocrea, G. Pilato, and G. M. Grasso, "Explainable AI at work! What can it do for smart agriculture?" in *Proc. IEEE 8th Int. Conf. Multimedia Big Data (BigMM)*, Dec. 2022, pp. 87–93.

[21] N. Almakayeel, S. Desai, S. Alghamdi, and M. R. N. M. Qureshi, "Smart agent system for cyber nano-manufacturing in Industry 4.0," *Appl. Sci.*, vol. 12, no. 12, p. 6143, Jun. 2022.

[22] H. Elhoone, T. Zhang, M. Anwar, and S. Desai, "Cyber-based design for additive manufacturing using artificial neural networks for Industry 4.0," *Int. J. Prod. Res.*, vol. 58, no. 9, pp. 2841–2861, May 2020.

[23] M. Ogunsanya and S. Desai, "Physics-based and data-driven modeling for biomanufacturing 4.0," *Manuf. Lett.*, vol. 36, pp. 91–95, Jul. 2023.

[24] E. K. Došilović, M. Breic, and N. Hlupic, "Explainable artificial intelligence: A survey," in *Proc. 41st Int. Conv. Inf. Commun. Technol., Electron. Microelectron. (MIPRO)*, May 2018, pp. 0210–0215.

[25] D. Weyns, J. Andersson, M. Caporuscio, F. Flammini, A. Kerren, and W. Löwe, "A research agenda for smarter cyber-physical systems," *J. Integr. Design Process Sci.*, vol. 25, no. 2, pp. 27–47, Aug. 2021.

[26] M. C. Zizic, M. Mladineo, N. Gjeldum, and L. Celent, "From Industry 4.0 towards Industry 5.0: A review and analysis of paradigm shift for the people, organization and technology," *Energies*, vol. 15, no. 14, p. 5221, Jul. 2022.

[27] K. Amarasinghe, "Explainable neural networks based anomaly detection for cyber-physical systems," Ph.D. dissertation, Dept. Comput. Sci., Virginia Commonwealth Univ., Richmond, VA, USA, 2019.

[28] A. B. Bakker, E. Demerouti, and A. I. Sanz-Vergel, "Burnout and work engagement: The JD-R approach," *Annu. Rev. Organizational Psychol. Organizational Behav.*, vol. 1, no. 1, pp. 389–411, Mar. 2014.